

EXPERIMENT – 01

AIM – WAP to implement bubble sort.

. PROGRAM :-

```
C bubbleSort.c > main()
1  #include <stdio.h>
2
3  void bubbleSort(int arr[], int n) {
4      for (int i = 0; i < n - 1; i++) {
5          int swapped = 0;
6
7          for (int j = 0; j < n - i - 1; j++) {
8              if (arr[j] > arr[j + 1]) {
9                  int temp = arr[j];
10                 arr[j] = arr[j + 1];
11                 arr[j + 1] = temp;
12             }
13             swapped = 1;
14         }
15     }
16
17     if (swapped == 0) {
18         break;
19     }
20 }
21
22
23 int main() {
24     int n;
25     printf("Enter number of elements: ");
26     scanf("%d", &n);
27
28     int arr[n];
29     printf("Enter %d elements: ", n);
30     for (int i = 0; i < n; i++) {
31         scanf("%d", &arr[i]);
32     }
33
34     bubbleSort(arr, n);
35
36     printf("Sorted array: ");
37     for (int i = 0; i < n; i++) {
38         printf("%d ", arr[i]);
39     }
40     printf("\n");
41
42     return 0;
43 }
44
```

OUTPUT :-

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\Vatsal\Vatsal.c> cd "c:\Users\Vatsal\Vatsal.c\" ;
Enter number of elements: 4
Enter 4 elements: 4
8
1
5
Sorted array: 1 4 5 8
PS C:\Users\Vatsal\Vatsal.c> |
```

EXPERIMENT – 02

AIM – WAP to implement Selection sort.

. PROGRAM :-

```
C selectionSort.c > main()
1  #include <stdio.h>
2
3  void selectionSort(int arr[], int n) {
4      for (int i = 0; i < n - 1; i++) {
5          int minIndex = i;
6          for (int j = i + 1; j < n; j++) {
7              if (arr[j] < arr[minIndex]) {
8                  minIndex = j;
9              }
10         }
11         if (minIndex != i) {
12             int temp = arr[i];
13             arr[i] = arr[minIndex];
14             arr[minIndex] = temp;
15         }
16     }
17 }
18
19 void printArray(int arr[], int n) {
20     for (int i = 0; i < n; i++) {
21         printf("%d ", arr[i]);
22     }
23     printf("\n");
24 }
25
26 int main() {
27     int n;
28     printf("Enter the number of elements: ");
29     scanf("%d", &n);
30
31     int arr[n];
32     printf("Enter %d elements:\n", n);
33     for (int i = 0; i < n; i++) {
34         scanf("%d", &arr[i]);
35     }
36
37     printf("Original array: ");
38     printArray(arr, n);
39
40     selectionSort(arr, n);
41
42     printf("Sorted array: ");
43     printArray(arr, n);
44
45     return 0;
46 }
47
```

OUTPUT :-

```
PS C:\Users\Vatsal\Vatsal.c> cd "c:\Users\Vatsal\Vatsal.c\"
Enter the number of elements: 4
Enter 4 elements:
2
1
4
7
Original array: 2 1 4 7
Sorted array: 1 2 4 7
PS C:\Users\Vatsal\Vatsal.c> cd "c:\Users\Vatsal\Vatsal.c\"
```

EXPERIMENT – 03

AIM – WAP to implement Insertion sort.

. PROGRAM :-

```
C insertionSort.c > main()
1  #include <stdio.h>
2
3  void insertionSort(int arr[], int n) {
4      for (int i = 1; i < n; i++) {
5          int key = arr[i];
6          int j = i - 1;
7
8          while (j >= 0 && arr[j] > key) {
9              arr[j + 1] = arr[j];
10             j--;
11         }
12         arr[j + 1] = key;
13     }
14 }
15
16 int main() {
17     int n;
18     printf("Enter number of elements: ");
19     scanf("%d", &n);
20
21     int arr[n];
22     printf("Enter %d elements: ", n);
23     for (int i = 0; i < n; i++) {
24         scanf("%d", &arr[i]);
25     }
26
27     insertionSort(arr, n);
28
29     printf("Sorted array: ");
30     for (int i = 0; i < n; i++) {
31         printf("%d ", arr[i]);
32     }
33     printf("\n");
34
35     return 0;
36 }
37
```

OUTPUT :-

```
PS C:\Users\Vatsal\Vatsal.c> cd "c:\Users\Vatsal\Vatsal.c\"
Enter number of elements: 4
Enter 4 elements: 2
3
6
1
Sorted array: 1 2 3 6
PS C:\Users\Vatsal\Vatsal.c>
```

EXPERIMENT – 04

AIM – WAP to implement Linear Search

. PROGRAM :-

```
C linearSearch.c > ...
1  #include <stdio.h>
2
3  int linearSearch(int arr[], int n, int key) {
4      for (int i = 0; i < n; i++) {
5          if (arr[i] == key) {
6              return i; // return index if found
7          }
8      }
9      return -1; // not found
10 }
11
12 int main() {
13     int n, key;
14     printf("Enter number of elements: ");
15     scanf("%d", &n);
16
17     int arr[n];
18     printf("Enter %d elements: ", n);
19     for (int i = 0; i < n; i++) {
20         scanf("%d", &arr[i]);
21     }
22
23     printf("Enter element to search: ");
24     scanf("%d", &key);
25
26     int result = linearSearch(arr, n, key);
27
28     if (result != -1) {
29         printf("Element found at index %d (position %d)\n", result, result + 1);
30     } else {
31         printf("Element not found in the array\n");
32     }
33
34     return 0;
35 }
36
```

OUTPUT :-

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\Vatsal\Vatsal.c> cd "c:\Users\Vatsal\Vatsal.c\" ;
Enter number of elements: 5
Enter 5 elements: 2
3
5
9
4
Enter element to search: 5
Element found at index 2 (position 3)
PS C:\Users\Vatsal\Vatsal.c>
```

EXPERIMENT 05

AIM – WAP to implement Binary Search

. PROGRAM :-

```
C binarySearch.c > main()
1  #include <stdio.h>
2
3  int binarySearch(int arr[], int n, int key) {
4      int left = 0, right = n - 1;
5
6      while (left <= right) {
7          int mid = left + (right - left) / 2;
8
9          if (arr[mid] == key) {
10             return mid;
11         }
12         else if (arr[mid] < key) {
13             left = mid + 1;
14         }
15         else {
16             right = mid - 1;
17         }
18     }
19
20     return -1; // not found
21 }
22
23 int main() {
24     int n, key;
25     printf("Enter number of elements (sorted array): ");
26     scanf("%d", &n);
27
28     int arr[n];
29     printf("Enter %d sorted elements: ", n);
30     for (int i = 0; i < n; i++) {
31         scanf("%d", &arr[i]);
32     }
33
34     printf("Enter element to search: ");
35     scanf("%d", &key);
36
37     int result = binarySearch(arr, n, key);
38
39     if (result != -1) {
40         printf("Element found at index %d (position %d)\n", result, result + 1);
41     } else {
42         printf("Element not found in the array\n");
43     }
44
45     return 0;
46 }
47
```

OUTPUT :-

```
PS C:\Users\Vatsal\Vatsal.c> cd "c:\Users\Vatsal\Vatsal.c\" ;
Enter number of elements (sorted array): 5
Enter 5 sorted elements: 1
3
5
7
9
Enter element to search: 7
Element found at index 3 (position 4)
PS C:\Users\Vatsal\Vatsal.c> █
```